

Comparative Study of Deep learning algorithms in the field of precision agriculture

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Abstract Corn is a mass-produced agricultural product that plays a major role in the food chain and many agricultural products in addition to biofuels. Furthermore, households in poor countries may depend on small-scale corn cultivation for their basic needs. However, corn crops are vulnerable to diseases, which greatly affects farming yields. Seeing the development of corn that continues to increase, it becomes essential to actively prevent common leaf diseases to increase production. In this work, we present a comparison of recent deep convolutional neural networks (CNN) based on transfer learning for automatic binary classification of corn leaf disease (healthy or unhealthy) at first, after we extend the study to the classification of three corn leaf diseases (i.e. Cercospora leaf spot, common rust, and northern leaf blight) in addition to the healthy plants, based on fine-tuned version of **VGG16**, **VGG19**, **DenseNet201**, **Inception_ResNet_V2**, **Inception_V3**, **Resnet50**, **MobileNet_V2** and **Xception**. By using an open dataset containing 4188 images (1306 common rust, 574 Cercospora leaf spot, 1146 northern leaf blight, 1162 healthy). As result we can conclude that the fine-tuned version of **DenseNet201** showed high performance in the binary classification task, on the hand **Inception_ResNet_V2** achieved the best performance on the multiclass classification task.

Keywords Image classification · Deep Learning algorithms · Pre-processing methods · Corn leaf disease · Leaf spot · Rust · Blight · Transfer Learning · Deep learning

1. Introduction

Plant diseases are one of the major factors in the destruction of agricultural crops, with an estimated global average yield loss of 2000 billion US dollars each year. Plants are adversely affected by infection, which may be fungal, viral, or bacterial. These diseases attack leaves, fruits, or branches and may appear in the form of changes in color, shape, or edges. Thus, it is imperative to investigate possible fast detection, control, and treatment options. Traditional naked eye inspection is the first line of defense toward disease diagnosis, but it is prone to errors. Furthermore, farmers can employ the wrong type of treatment method or chemical. It generally needs a plant pathologist to analyze and discover the type of infection, which may be costly, not prompt enough, or unavailable.

Corn is a versatile crop with various uses. It serves as a staple food for many populations, providing a significant source of calories, carbohydrates, and essential nutrients. Beyond its role as a dietary staple, corn is used in a wide range of products and industries, including animal feed, ethanol production, pharmaceuticals, and bio-based materials. However, this crop is susceptible to various diseases that can have devastating impacts on its growth and yield. Common rust, blight, and gray leaf spot are three significant diseases that afflict corn plants. Common rust is caused by the fungus *Puccinia sorghi*, manifests as reddish-brown pustules on corn leaves, stems, and husks. It can lead to reduced photosynthesis, stunted growth, and ultimately, yield losses. Blight is caused by *Exserohilum turcicum* fungus, results in elliptical lesions with yellow halos on the leaves, leading to premature drying and death of plant tissues. This disease can severely impact corn production, especially during periods of high humidity and warm temperatures. Gray leaf spot, caused by two types of fungi *Cercospora zeae-maydis* and *Cercospora Zeina*, is characterized by grayish lesions on the leaves, which can expand and merge, causing defoliation and reduced grain fill.

Artificial intelligence (AI) and deep learning algorithms have revolutionized the field of agriculture, particularly in the detection and diagnosis of corn leaf diseases. By leveraging large datasets and complex computational models, these technologies have enabled accurate and timely identification of various diseases that affect corn plants. Deep learning algorithms are trained on vast amounts of labeled images of healthy and diseased corn leaves, allowing them to learn intricate patterns and features associated with specific diseases. This training enables the algorithms to classify new images of corn leaves and detect signs of disease with remarkable precision. Through the use of AI, farmers and researchers can identify and respond to corn leaf diseases, by implementing targeted interventions to prevent further spread and minimize crop losses. Motivated by the fastest and accurate detection rate of corn leaf diseases using DL, our work will present a comparison of 8 recent deep learning convolution neural network architecture for automatic detection of maize leaf images in order to answer the following research questions :

(RQ1) : What is (are) DL algorithm(s) that performs well on both binary and multiclass problems?

(RQ2) : Does a high performing DL algorithm trained on a binary classification problem means that it will automatically achieve high performances on the multiclass problem as well ?

Our paper's contribution are as follows:

(1) We implemented fine tuned version of VGG16, VGG19, DenseNet201, Inception_ResNet_V2, Inception_V3, Resnet50, MobileNet_V2 and Xception [1, 2].

(2) the various models have been tested on Maize Leaf Disease dataset [3, 4] for both binary and multiclass classification extending the work of current state-of-art algorithms that focus only on the multiclass classification.

The reminder of this paper is organized as follows. Section 2 stats some related works. In Section 3 we introduce our proposed contribution. Section 4 presents the results obtained in this study as well as interpretations to each result. In section 5 we discuss the results found in section 4 . The conclusions were kept for the last section.

2. Related works

Recently, several research works in AI-based corn disease detection and classification were conducted. Padilla et al. [5] designed a system using Raspberry Pi to capture and process corn leaf images and classify the type of disease exhibited by those leaves. They used CNN implemented using the multiprocessing programming library OpenMP. The highest accuracy that they reported was 93% for the classification of leaf blight. However, the rust and leaf spot diseases were detected with an 89% accuracy. Xu et al. [6] modified an AlexNet CNN by adding a new convolution layer and an inception module. They compared their model with transfer learning based on VGGNet-16, DenseNet, ResNet-50 and AlexNet. Their modified model achieved a 93.28% accuracy.

3. Proposed Contribution

Deep learning methods have recently demonstrated huge potential with state-of-the art performance on image processing and computer vision [7]. These algorithms have been applied in various tasks of precision agriculture with high performance in segmentation, detection, and classification.

Following the success of deep learning in plant leaf image processing, our work is going to deeply compare different fine-tuned architectures (VGG16, VGG19, DenseNet201, Inception_ResNet_V2, Inception_V3, Resnet50, MobileNet_V2 and Xception). The rest of this section will be a detail of the proposed DL models.

Deep learning architectures have found applications in several fields, such as healthcare [1, 2], finance, agriculture, and robotics. The most investigated DL models are VGG16, VGG19, DenseNet201,

Inception_ResNet_V2, Inception_V3, Resnet50, MobileNet_V2 and Xception, we chosen these 8 models due to the accuracies they offer regardless of the classification task they are applied on.

- **VGG16 and VGG19**

VGG16 and VGG19 are deep convolutional neural network architectures that have made significant contributions to the field of computer vision. They were developed by the Visual Geometry Group (VGG) at the University of Oxford [8]. These architectures are characterized by their stacked arrangement of convolutional layers, followed by fully connected layers. VGG16 consists of 16 convolutional layers, while VGG19 has 19 convolutional layers. Both architectures use 3×3 convolutional filters with a stride of 1 and employ max-pooling layers to reduce spatial dimensions.

The key idea behind VGG16 and VGG19 is to use a series of simple convolutional operations to learn complex hierarchical features from input images. By increasing the depth of the network, VGG models are able to capture more intricate patterns and achieve better representation learning. The deep layer structure enables them to learn rich and abstract features, which makes them highly effective in image classification tasks.

- **Inception_V3**

Inception_V3 is a popular deep learning architecture developed by Google researchers [9]. It's known for its efficient use of computational resources and high accuracy. The architecture is based on the concept of "inception modules," which consist of multiple parallel convolutional layers with different filter sizes. These parallel branches allow the network to capture features at various scales and levels of abstraction simultaneously. Inception-V3 also incorporates techniques like batch normalization and factorized convolutions to further improve efficiency and reduce overfitting. This model accepts an input image size of 299×299 pixels.

The model has been widely used in applications such as visual recognition, image captioning, and fine-grained categorization. Its exceptional performance and versatility have made Inception-V3 a valuable tool for researchers and practitioners in the field of computer vision.

- **ResNet50**

ResNet50 is a deep convolutional neural network architecture that has gained significant attention and popularity in the field of computer vision. It is a variant of the ResNet (Residual Network) family [10], designed to overcome the problem of degradation when training very deep neural networks. the key innovation of ResNet50 lies in the introduction of residual blocks, which enable the network to learn residual mappings instead of directly learning the underlying mappings. These residual blocks contain skip connections, allowing information to bypass certain layers and flow directly to deeper layers. This addresses the vanishing gradient problem and facilitates the training of much deeper networks.

ResNet50 consists of 50 layers, including convolutional layers, pooling layers, and fully connected layers. It incorporates bottleneck structures in the residual blocks, which reduce the computational complexity while maintaining model performance. This design choice allows ResNet50 to achieve a good balance between accuracy and efficiency. The model accepts image of size 224×224 pixels.

- **Inception_ResNet_V2**

Inception_ResNet_V2 is a convolutional neural network trained on more than a million images from the ImageNet dataset [11].It is an architecture that combines the Inception module and residual

connections to achieve remarkable performance in image recognition tasks. It is an extension of the original Inception architecture and builds upon the idea of using multiple parallel convolutions at different scales to capture rich spatial information. the inception structure and the residual connection. The model accepts images of 299×299 image, and its output is a list of estimated class probabilities. The architecture has been extensively used in computer vision tasks, such as image classification, object detection, and segmentation. Inception-ResNet-V2 has achieved state-of-the-art results on benchmark datasets and has been utilized in real-world applications, including autonomous driving, medical imaging, and visual recognition systems.

- **DenseNet201**

DenseNet201 is a deep learning architecture that belongs to the family of DenseNets [12], which are known for their dense connectivity patterns. DenseNet201 is specifically designed for image classification tasks and has gained popularity due to its superior performance and efficient parameter usage. The architecture of DenseNet201 accepts an input image of 224×224 pixels and consists of several densely connected blocks, where each block connects all preceding layers directly to its subsequent layers. This dense connectivity ensures that information from earlier layers is preserved and propagated throughout the network, promoting feature reuse and enhancing gradient flow during training. By leveraging this connectivity pattern, DenseNet201 achieves strong feature representation while mitigating the vanishing gradient problem.

- **MobileNet_V2**

MobileNet_V2 [13] is a convolutional neural network being an extension of the original MobileNet architecture aiming to provide better performance and accuracy while maintaining low computational and memory requirements. It is made of only 54 layers and has an input image size of 224×224 . Its main characteristic is the use of depthwise separable convolutions, which split the standard convolution operation into two separate layers: a depthwise convolution that performs filtering on each input channel independently, followed by a pointwise convolution that combines the filtered outputs. This technique significantly reduces the number of parameters and computations compared to traditional convolutions, making MobileNetV2 faster and more lightweight.

MobileNetV2 has been widely adopted for various computer vision tasks on mobile and resource-constrained devices. Its compact size and efficient design make it well-suited for real-time applications like object detection, image classification, and semantic segmentation on mobile platforms. The architecture strikes a good balance between model size, computational complexity, and accuracy, enabling it to deliver impressive results even on devices with limited computational power and memory.

- **Xception**

Xception is deep learning architecture presented by Chollet in 2016 [14], It represents a significant advancement in convolutional neural networks (CNNs) and has gained attention for its exceptional performance in image classification tasks. Xception is 71 layers deep and it's based on the concept of the Inception module, which aims to capture both local and global information within an image. However, what sets Xception apart is its unique design that employs depthwise separable convolutions. It showed good results compared to VGG16, Resnet and Inception in classical classification problems. Xception has an input image size of 299×299 .

Table 1 shows the number of parameters as calculated from the pre-trained models, and the required input size for each one.

Table 1: the depth in term of number of layers, and the required input size for each model

Model	Depth (Layers)	Input Size
MobileNet_V2	54	224 x 244
ResNet50	50	224 x 244
DenseNet201	201	224 x 244
VGG16	16	224 x 244
VGG19	19	224 x 244
Inception_V3	48	299 x 299
Inception_ResNet_V2	164	299 x 299
Xception	71	299 x 299

4. Experimental Results and Analysis

We compared all the models mentioned above after they have been fine tuned for the automatic binary and multiclass classification on the public image dataset of corn leaf disease [3, 4]. A pipeline of preprocessing methods was applied to the images of our dataset consisting of normalization, then CLAHE, followed by data augmentation in order to enhance their quality and improve the generalizability of our models.

4.1. Dataset

The present work introduces the publicly available dataset of maize leaf disease [15], this dataset has been crafted by combining the popular PlantVillage dataset [3], and the PlantDoc dataset [4], certain images were excluded in the process as they were considered useless.

The resulting dataset is composed of 4188 images divided into 4 classes, each representing one of four possible disease states, as follows: 574 Cercospora leaf spot, 1306 common rust, 1146 northern leaf blight, 1162 healthy.

Samples of the images of the three diseases and healthy leaves are shown in Figure 1.

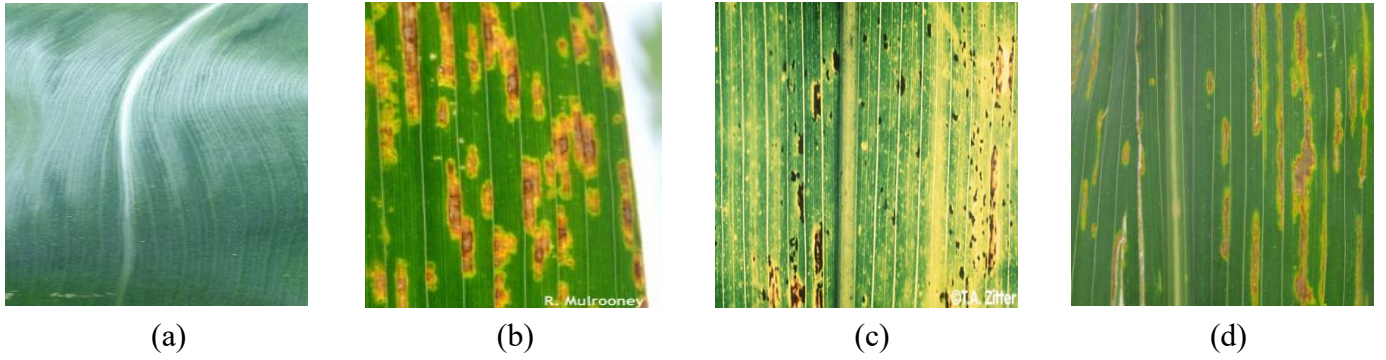


Figure 1: Sample images from the three disease classes and the healthy one: (a) healthy. (b) Grey. Leaf Spot. (c) Common Rust. (d) Blight.

4.2. Data pre-processing and splitting

Data preprocessing and splitting are essential steps in the deep learning pipeline. The motivation behind image pre-processing is to improve each image's quality by eliminating (or decreasing) the noise associated with each image, increasing the contrast, and deleting low or high frequencies. In this study, we used intensity normalization and Contrast Limited Adoptive Histogram Equalization (CLAHE) as two preprocessing techniques.

Intensity Normalization is a crucial step in image processing and analysis tasks, it aims to standardize the intensity levels across different images which helps to speed up the convergence of the model.

In this paper, input images (fig 2(a)) are normalized to the standard normal distribution using min-max normalization (Eq 1), which is computed as :

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

Where x is the pixel intensity, x_{min} and x_{max} are minimum and maximum intensity values of the input image.

Moreover, before feeding input images to the proposed algorithms, we applied CLAHE (Fig 2(c)) to improve the contrast of images in hand. Figure 2 shows an example of the use of normalization and CLAHE.

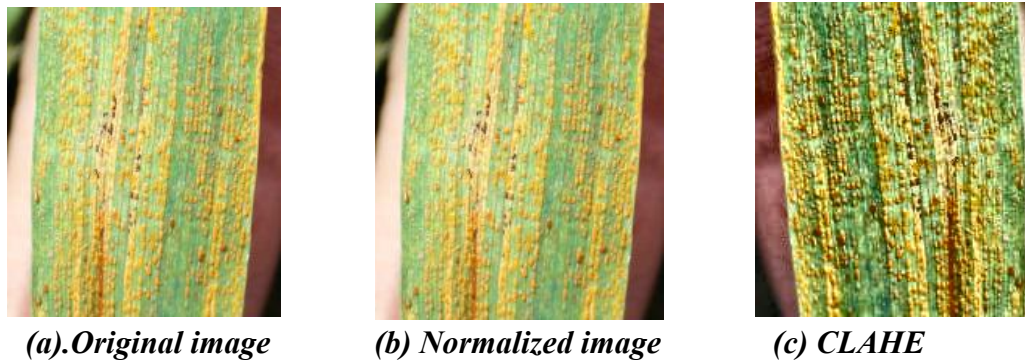


Figure 2: maize leaf image pre-processing.

Concerning data splitting , 80% of images were used for training, the remaining 20% were kept for testing. The previous splitting percentages were respected in both binary and multiclass tasks, also, we ensured the use of a random process to split the original dataset to avoid statistical bias.

Finally, to overcome the observed problem of imbalanced classes, Data augmentation was employed as a last step to oversample minority classes, resulting a clean dataset to train our models.

4.3. Data augmentation

Data augmentation is a crucial technique used after data preprocessing and splitting to increase the diversity and quantity of training data. By applying various transformations and modifications to the existing dataset, data augmentation helps in improving the generalization and robustness of models. The techniques of data augmentation we used include transformations such as flips, rotations, shears, zooms (Table 2)

4.4. Training and classification

Following data pre-processing, splitting, and data augmentation techniques, our training dataset has been augmented and prepared for the subsequent feature extraction step using the proposed models. The purpose of this step is to extract relevant and meaningful features from the data. Each proposed model extracts features from the input images, and these features are then flattened and combined to create vectorized feature maps. The resulting feature vector is subsequently fed into a multilayer perceptron, which performs the classification task by assigning each image to its corresponding class. To assess the performance of our approach, we evaluate it on a separate set of test images using the trained model.

Tableau 2: data augmentation techniques used

Argument	Parameter value	Description
Rescale	1/255.0	Scale images from 0-255 to floats 0-1
Rotation range	90	Degree of the random rotations
Horizontal and Vertical shift range	0.2	The parameter value of horizontal and vertical shifts (20%) is a fraction of the given dimension
Shear range	0.2	Controls the angle in counter
Zoom range	0.2	Permet à l'image d'être "zoomée" ou 'dézoomée'
Horizontal flip	True	Contrôle quand une entrée donnée est autorisée à être retournée horizontalement pendant le processus d'entraînement
Fill mode	Nearest	Il s'agit de l'option par défaut où la valeur de pixel la plus proche est répétée pour toutes les valeurs vides.

4.5. Experimental setup

Towards an automatic classification of corn leaf disease, our experiments were conducted using the following parameters: all the images in our dataset were resized to 224×224 pixels, except for the **Inception_V3**, **Inception_Resnet_V2**, and **Xception** models, where they were resized to 299×299 . For model training, we set the batch size to 32 with the number of epochs set to 250. Regarding the training and test sets, we used 175 and 41 samples, respectively, for binary classification, while for multi-class classification, we used 132 training samples and 31 test samples. We chose to use the Adam algorithm with $\beta_1=0.9$ and $\beta_2=0.999$ for optimization. The learning rate was initially set to 0.0001 and then reduced to 0.000001 during training. Additionally, to avoid model overfitting, we applied weight decay and L2 regularization.

For all models, we trained fully connected layers using the ReLU activation function. Each layer was followed by a dropout layer with a probability of 0.5, which helped reduce overfitting. We also adapted the last dense layer of each model to produce two classes for binary classification or four classes for multi-class classification, instead of the 1000 classes used by ImageNet.

Finally, our models were trained using virtual machines offered by public cloud providers (Azue, Google cloud, and Oracle cloud) running OS Ubuntu Server 22.04.

4.6. Evaluation metrics

The metrics used to evaluate the performance of the CNN models are shown in Equations (2)–(6). Sensitivity (SEN) measures the ability of the model to identify a leaf image as belonging to the correct disease class out of all the positive images, which is affected by the existence of false negatives. Moreover, Specificity (SPE) measures the model's ability to predict true negatives of each available category. Precision (PRE) measures the ratio of false positives to all cases identified as positive (i.e., false positives included). The Accuracy (ACC) measures the ratio of the sum of true positives and true negatives to the total number of testing images. F1 score (F1) is considered a more reliable measure of the model classification performance. These metrics are defined as follows:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (2)$$

$$SEN = \frac{TP}{TP + FN} \times 100 \quad (3)$$

$$SPE = \frac{TN}{TN + FP} \times 100 \quad (4)$$

$$PRE = \frac{TP}{TP + FP} \times 100 \quad (5)$$

$$F1 = 2 \times \frac{Recall \times Precision}{Recall + Precision} \times 100 \quad (6)$$

where: **TP** stands for: True Positive. **FP**: False Positive. **TN**: True Negative, and **FN**: False Negative.

4.7. Preliminary study of the results

This section presents the results of the binary and multiclass classification for maize leaf disease dataset [15] after fine tuning eight architectures deep learning (**VGG16**, **VGG19**, **DenseNet201**, **Inception_ResNet_V2**, **Inception_V3**, **Resnet50**, **MobileNet_V2** and **Xception**). The results include the curves of accuracy and loss for both training and testing sets, along with the confusion matrix.

4.7.1. binary classification results

This subsection presents the results of the binary classification of maize leaf disease []. Before diving into the results, let's first define some parameters related to the deep learning process: The training curve is derived from the training dataset and gives an indication of the model's learning progress. On the other hand, the testing curve is generated from a separate testing dataset and evaluates how well the model generalizes. The training and testing loss, unlike accuracy, are calculated by summing up the errors made for each example in their respective datasets. Additionally, the confusion matrix provides a detailed representation of the classification results for the images.

• VGG16

According to the Fig. 3, it is observed that the accuracy curve of training data is rapidly increasing from epoch 0 to epoch 5 where the accuracy is equal to 96.28%, after that, it begins to increase slightly until epoch 250 where the accuracy is equal to 97.67%. The same applies to the accuracy curve of testing data with an accuracy of 96.26% for epoch 250.

Concerning the loss curve of training data, it is rapidly decreasing from epoch 0 to epoch 5 where the loss is 1.45. At that point, it begins to decrease slightly until the end of training (epoch 250) where the loss is equivalent to 0.19. Same for loss curve of testing data with a loss of 0.21 for epoch 250.

From the confusion matrix, it is noted that the first class (healthy class), the model recognizes 653 images correctly, but 6 were marked as unhealthy. Likewise, for the second image's class (unhealthy), the model was capable to identify 614 images correctly, unlike 43 images were marked as healthy.

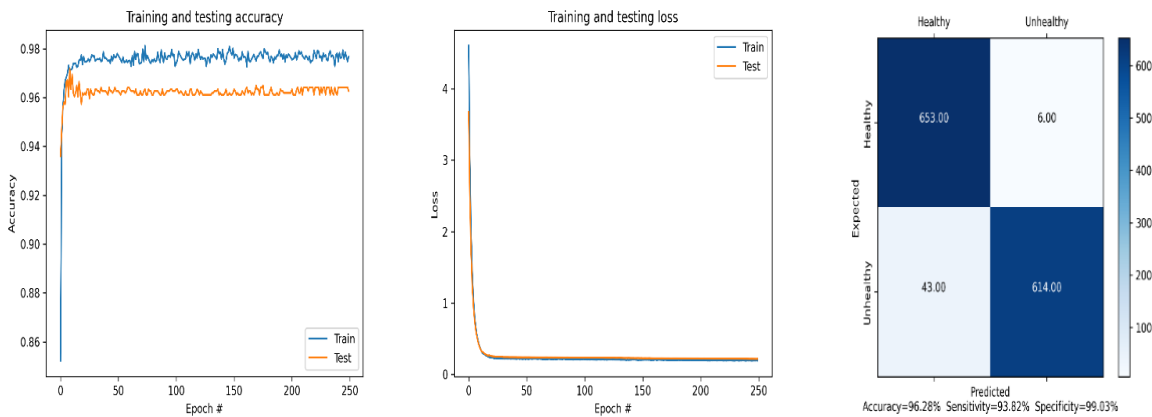


Figure 3: Accuracy, loss, and confusion matrix of VGG16.

• VGG19

According to Fig 4, we observe that the training accuracy increases rapidly from epoch 0 to 10, reaching a value of 96.56%. Afterward, it continues to increase slowly until epoch 250, where its value is 97.33%. The same pattern applies to the testing accuracy, which reaches 96.26% at epoch 250.

Regarding the training loss curve, it decreases rapidly from epoch 0 to 10, reaching a value of 0.49. From this point, it continues to decrease slowly until epoch 250, where its value equals 0.25. The same behavior is observed for the testing loss, which equals 0.21 at epoch 250.

Analyzing the confusion matrix, it is noted that in the first class (corn Healthy), the model correctly classified 655 images as corn healthy, but 4 images were misclassified as corn unhealthy. Similarly, in the second class (corn unhealthy), the model correctly classified 601 images, but 52 images were misclassified.

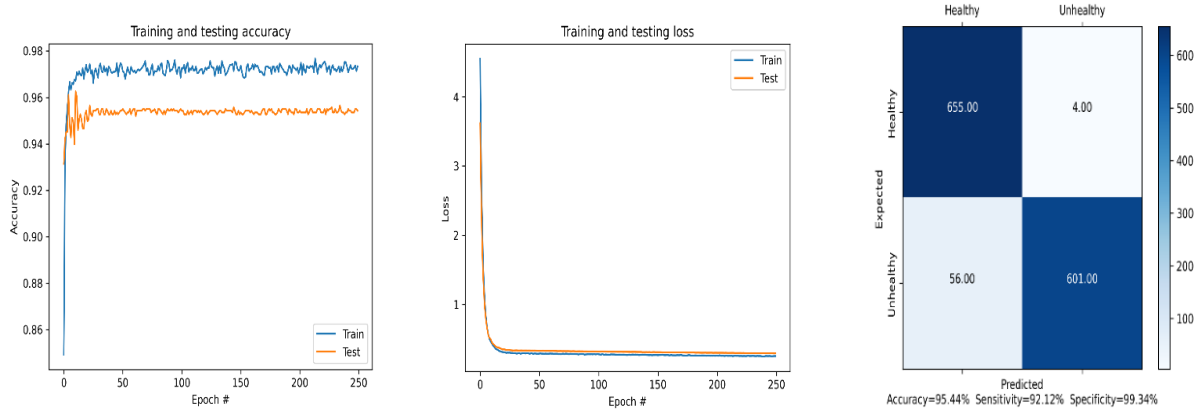


Figure 4: Accuracy, loss, and confusion matrix of VGG19.

• Inception_V3

According to Figure 5, we observe that the training accuracy increases rapidly from epoch 0 to 10, reaching a value of 96.56%. Afterward, it continues to increase slowly until epoch 250, where its value is 97.33%. The same pattern applies to the testing accuracy, which reaches 96.26% at epoch 250. Regarding the training loss curve, it decreases rapidly from epoch 0 to 10, reaching a value of 0.49. From this point, it continues to decrease slowly until epoch 250, where its value equals 0.25. The same behavior is observed for the testing loss, which equals 0.21 at epoch 250. Analyzing the confusion matrix, it is noted that in the first class (corn Healthy), the model correctly classified 653 images as corn healthy, but 6 images were misclassified as corn unhealthy. Similarly, in the second class (corn unhealthy), the model correctly classified 657 images, but 0 images were misclassified.

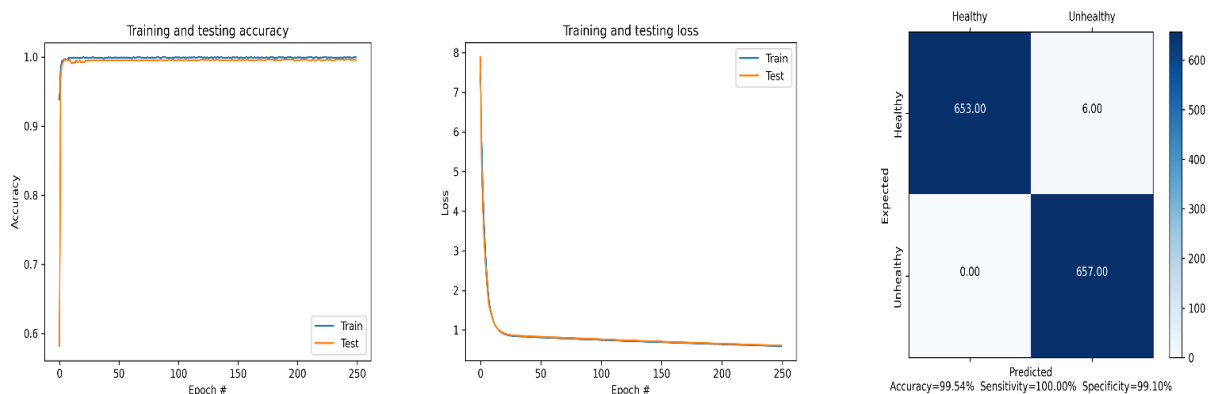


Figure 5: Accuracy, loss, and confusion matrix of Inception_V3.

• ResNet50

Figure 6 illustrates the results obtained by ResNet50. In fact, the training and testing accuracy exhibit exponential growth between epochs 0 and 20, reaching values of 99.98% and 99.84% respectively. After this point, the values begin to stabilize, with 99.96% for training and 99.84% for testing.

Regarding the loss curve, we observe a perfect overlap between the training and testing curves. Between epochs 0 and 21, the loss rapidly decreased to a value of 0.09 before stabilizing at 0.04 in epoch 250. The confusion matrix indicates that for images in the first class (corn Healthy), 657 images were correctly classified. However, 2 images were misclassified as corn unhealthy. In the second class (corn unhealthy), all images were correctly classified into the corresponding state.

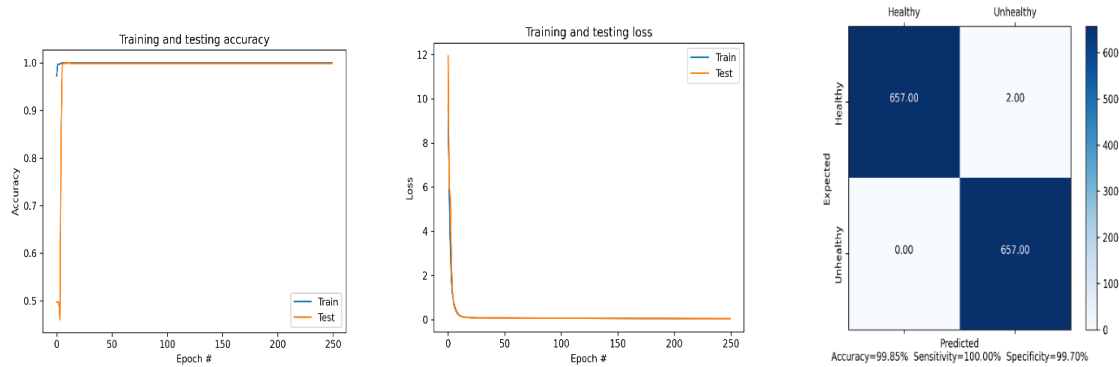


Figure 6: Accuracy, loss, and confusion matrix of ResNet50.

• Inception_ResNet_V2

Figure 7 shows that from epoch 0 to epoch 15, both the training and testing accuracy curves increase steadily until reaching a value of 99.74%. After epoch 15, the accuracy values begin to stabilize, with 99.98% for training and 99.69% for testing. As for the loss curve, we observe an excellent overlap between the training and testing curves, where the loss values are 0.48 (at epoch 17) and 0.36 (at epoch 250).

For the first class (Healthy Corn), 655 images were correctly classified, while 4 images were misclassified as Unhealthy Corn. In the second class (Unhealthy Corn), all the images (657) were correctly classified in their corresponding state.

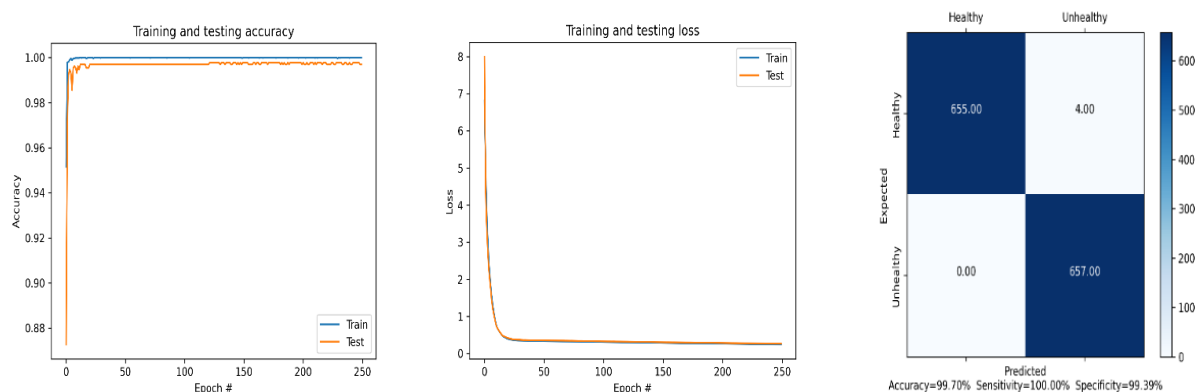


Figure 7: Accuracy, loss, and confusion matrix of Inception_ResNet_V2.

- **DenseNet201**

The accuracy curve obtained for the training data increases rapidly until reaching a value of 99.94% (Fig. 8). After epoch 18, the accuracy enters a stable phase, reaching respective values of 99.98% and 99.92% for the training and test data. We observe a perfect overlap of the loss curve for the training and test data. In fact, from epoch 0 to epoch 17, the loss decreases rapidly until reaching a value of 0.90, after which it stabilizes until epoch 250, where it equals 0.58. The confusion matrix shows that for the first set of images (healthy class), the model was able to correctly recognize all the images (659 images) of the healthy class. The model was also able to correctly identify all the images (657 images) of the second class (unhealthy class).

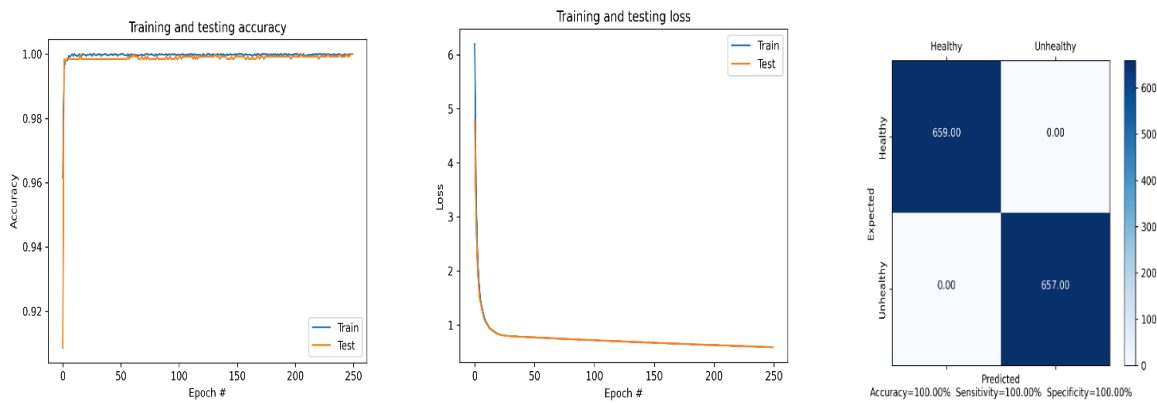


Figure 8: Accuracy, loss, and confusion matrix of DenseNet201.

- **MobileNet_V2**

As shown in Figure 9, we can observe that from epoch 0 to 7, the training and test accuracy increases until reaching an accuracy value of 99.96%. After epoch 7, the accuracy stabilizes and reaches 99.98% and 99.77% for the training and test data, respectively. Regarding the loss curve, we notice an excellent fit between training and testing loss. Until epoch 9, the loss value decreases rapidly to reach 0.05. Then it converges to 0.034. As for the confusion matrix, for the first images (healthy class), the model successfully identified 656 images correctly in the healthy class, but 3 were classified as unhealthy. Whereas, for the unhealthy class, all images (657 images) were correctly identified as corn unhealthy.

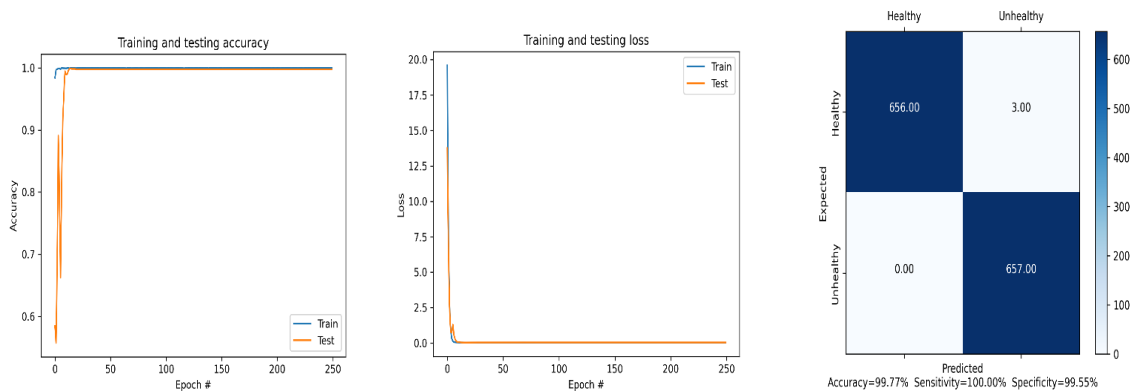


Figure 9: Accuracy, loss, and confusion matrix of MobileNet_V2.

- **Xception**

It is noted that the accuracy of the training data rapidly increases from epoch 0 to 18, reaching 99.92% (see Fig. 10). Afterward, it stabilizes until the end of training, with an accuracy of 99.96%. For the test data, we observe a rapid increase from epoch 0 to 12, where the accuracy reaches 99.61%. It then stabilizes at a value of 99.69% by epoch 250. Regarding the loss curve, the values decrease rapidly from epoch 0 to epoch 9, with a value of 0.03 for the training data and 0.047 for the test data. After epoch 9, the training loss converges to 0.027, and the test loss converges to 0.041. When examining the confusion matrix, we can see that for the first class of images (healthy class), the model correctly classified 655 images. Additionally, 4 images were identified as unhealthy. The model was also able to correctly classify all the images in the second class (unhealthy class).

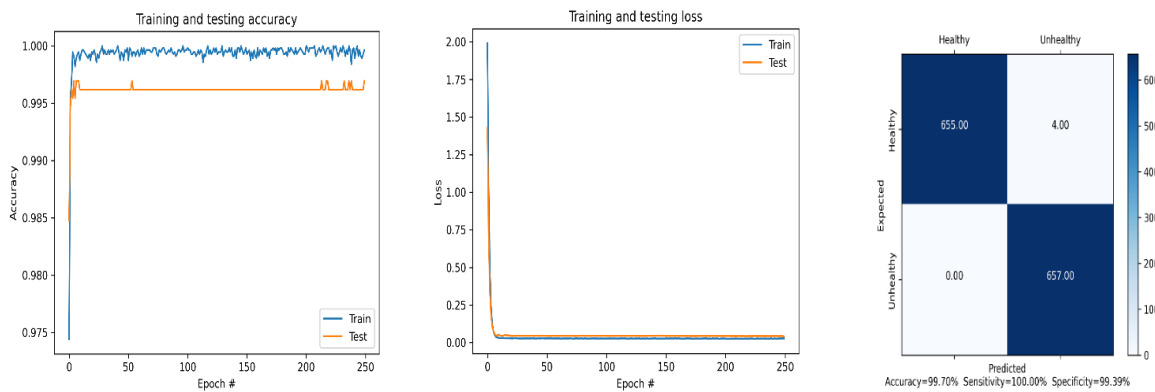


Figure 10: Accuracy, loss, and confusion matrix of Xception.

4.7.2. Multi-class classification results

This subsection presents the results of the multi-class classification of maize leaf disease. We use the same evaluation metrics as those used in the binary classification (see previous subsection).

- **VGG16**

In Figure 11, we observe an increase in training and testing accuracy from time 0 to 15, reaching 80.87% and 71.57% respectively. After epoch 15, the accuracy stabilizes at 82.16% for training and 72.58% for testing. The training loss curve shows rapid decrease until epoch 10, reaching 2.15, and then converging to 1.53.

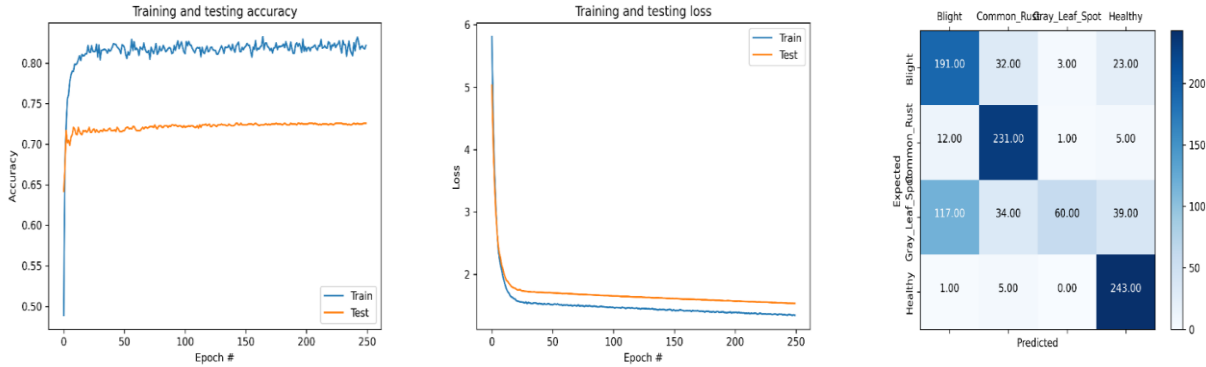


Figure 11: Accuracy, loss, and confusion matrix of VGG16.

Table 3 of VGG16 demonstrates good precision, sensitivity, and specificity for the Healthy class, with values of 78.38%, 97.59%, and 91.10% respectively. The Common Rust class exhibits equivalent sensitivity and specificity of 92.77% and 90.50%, with an acceptable precision of 76.06%. For the Blight class, precision, sensitivity, and specificity are 59.5%, 76.70%, and 82.62% respectively. However, the Gray Leaf Spot class shows low sensitivity of 24%, with high specificity (99.46%) and precision (93.75%). Accuracy values are 92.67%, 80.54%, and 80.54% for Healthy, Common Rust, and Gray Leaf Spot classes respectively.

Table 3: evaluation metrics table for VGG16

Classe	TP	TN	FN	FP	ACC(%)	SEN(%)	SPE(%)	PRE(%)	F1 Score (%)
Blight	191	618	58	130	81.14	76.70	82.62	59.5	67.01
Common Rust	231	677	18	71	91.10	92.77	90.50	76.94	83.84
Gray Leaf Spot	60	743	190	4	80.54	34	99/46	93.75	38.71
Healthy	243	681	6	67	92.67	97.59	91.10	76.38	86.93

- **VGG19:**

In Figure 12, training and testing accuracy increase from time 0 to 10, reaching 77.71% and 62.09% respectively. After epoch 15, accuracy stabilizes at 81.35% for training and 64.81% for testing. The training loss curve shows excellent fitting, decreasing rapidly until epoch 9 to 2.21 and converging to 1.54.

Table 4 of VGG19 indicates that the Healthy class had good precision, sensitivity, and specificity of 67.39%, 98.79%, and 84.09% respectively, with an accuracy of 87.76%. For the Common Rust class, sensitivity and specificity were 94.35% and 83.68% respectively, with an acceptable precision of

65.73%. The Blight class had reasonable precision, sensitivity, and specificity of 54.58%, 50.2%, and 86.09% respectively, with an accuracy of 77.13%. However, the Gray Leaf Spot class exhibited low sensitivity of 17.2%, with high specificity (99.08%) and precision (91.48%), resulting in an accuracy of 69.37%.

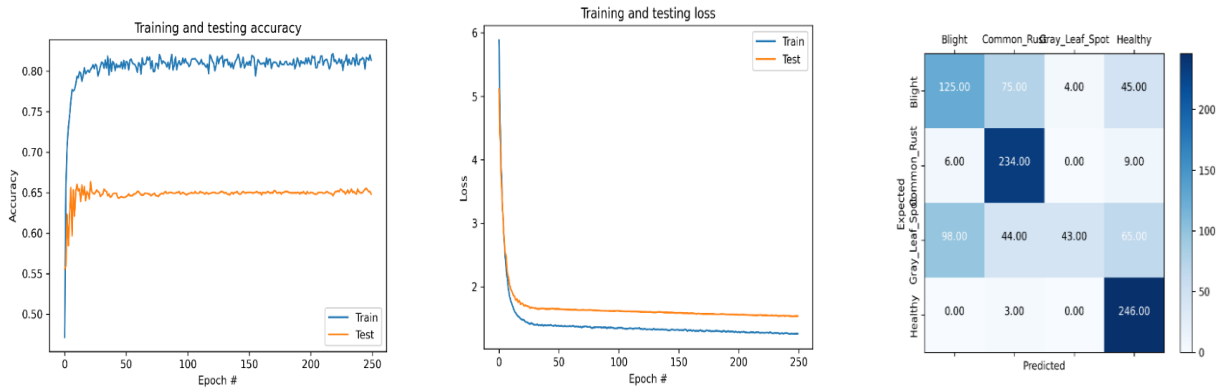


Figure 12: Accuracy, loss, and confusion matrix of VGG19.

Table 4: evaluation metrics table for VGG19

Classe	TP	TN	FN	FP	ACC(%)	SEN (%)	SPE (%)	PRE(%)	F1 Score (%)
Blight	125	644	124	104	77.13	50.20	86.09	54.58	52.29
Common Rust	234	629	14	122	86.24	94.35	83.68	65.73	77.48
Gray Leaf Spot	43	435	207	4	69.37	17.2	99.08	91.48	28.95
Healthy	246	629	3	119	87.76	98.79	84.09	67.39	80.12

• Inception_V3

In Figure 13, training and testing accuracy increase to 93.43% and 85.68% respectively until epoch 10, then stabilize at 98.03% and 89.41%. The training loss curve shows excellent fitting, decreasing rapidly until epoch 9 to 4.23 and converging to 2.21.

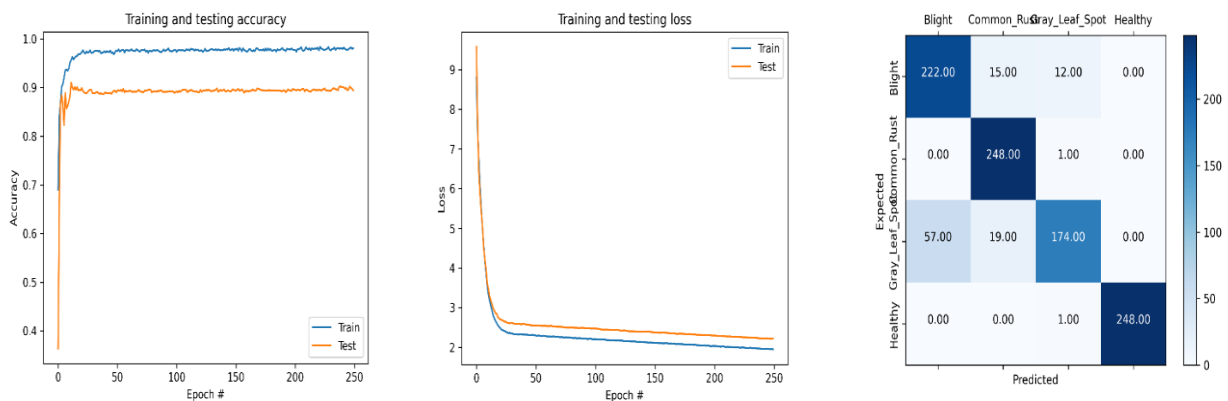


Figure 13: Accuracy, loss, and confusion matrix of Inception_V3.

Inception_V3 is the best-performing model, accurately classifying most images except for some in the Gray Leaf Spot class. Table 5 shows excellent precision, sensitivity, and specificity of 100%, 99.59%, and 100% respectively for the Healthy class with 99.89% accuracy. The Common Rust class has sensitivity and specificity of 99.59% and 95.45%, precision of 87.94%, and 96.48% accuracy. The Blight class shows reasonable precision and sensitivity of 79.56% and 89.15%, with 92.37% specificity and 91.57% accuracy. The Gray Leaf Spot class exhibits sensitivity of 69.6%, specificity of 98.12%, precision of 92.55%, and 90.97% accuracy.

Table 5: evaluation metrics table for Inception_V3

Classe	TP	TN	FN	FP	ACC(%)	SEN (%)	SPE (%)	PRE (%)	F1 Score (%)
Blight	222	691	27	57	91.57	89.15	92.37	79.56	84.08
Common Rust	248	714	1	34	96.48	99.59	95.45	87.94	93.40
Gray Leaf Spot	174	733	76	14	90.97	69.6	98.12	92.55	79.64
Healthy	248	748	1	0	99.89	99.59	100	100	99.74

• ResNet50

In Figure 14, training and testing accuracy increase until epoch 9 to 97.21% and 84.37%, then stabilize at 99.90% and 90.52%. The training loss curve shows excellent fitting, decreasing rapidly until epoch 9 to 3.97 and converging to 1.22.

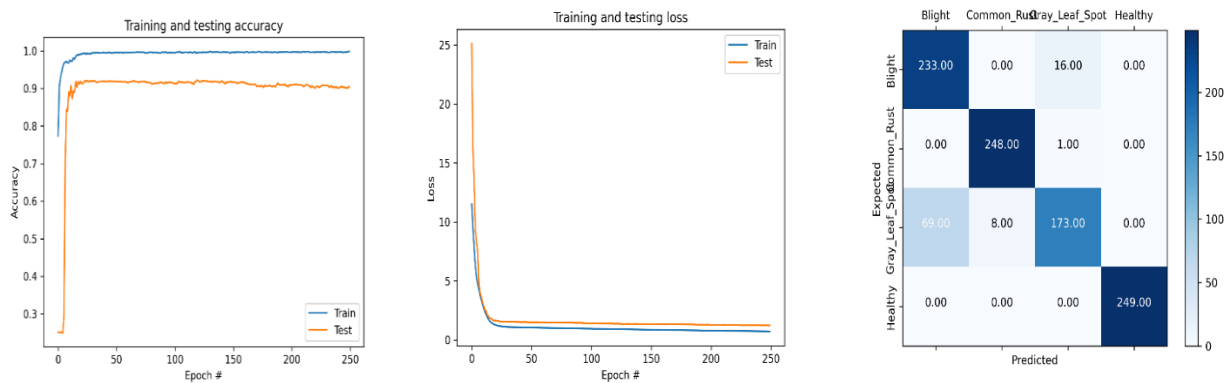


Figure 14: Accuracy, loss, and confusion matrix of Inception_V3.

The Healthy class (Table 6) is well-identified with 100% precision, sensitivity, and specificity, resulting in 100% accuracy. ResNet50 performs excellently for the Common Rust class with 96.87% precision, 99.59% sensitivity, and 98.83% specificity, achieving 99.09% accuracy. The Blight class shows reasonable precision, sensitivity of 93.57%, and specificity of 90.77%, with 77.15% precision and 91.47% accuracy. The Gray Leaf Spot class exhibits sensitivity of 69.2%, specificity of 97.72%, precision of 91.05%, and 90.57% accuracy, indicating relatively good detection compared to previous algorithms.

Table 6 : evaluation metrics table for ResNet50

Classe	TP	TN	FN	FP	ACC (%)	SEN (%)	SPE (%)	PRE(%)	F1 Score (%)
Blight	233	679	16	69	91.47	93.57	90.77	77.15	84.57
Common Rust	248	740	1	8	99.09	99.59	98.93	96.87	98.71
Gray Leaf Spot	173	73°	77	17	90.57	69.2	97.73	91.05	78.63
Healthy	249	748	0	0	100	100	100	100	100

• Inception_ResNet_V2

In Figure 14, training and testing accuracy increase until epoch 8 to 96.76% and 90.52%, then stabilize at 99.85% and 92.94%. The training loss curve shows excellent fitting, decreasing rapidly until epoch 8 to 3.83 and converging to 0.92.

The Inception_ResNet_V2 model is the most performing algorithm, achieving an accuracy of almost 93%, the highest among all algorithms used in this work.

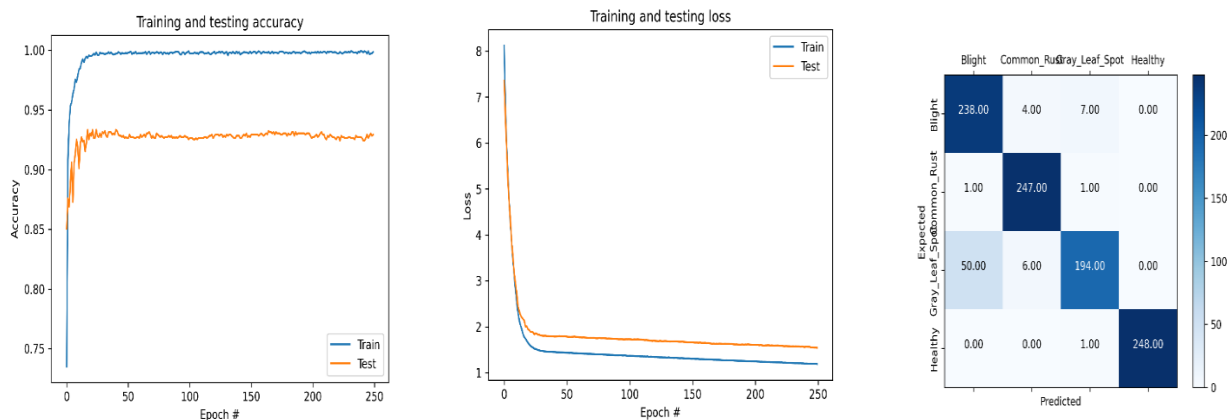


Figure 14 : Accuracy, loss, and confusion matrix of Inception_V3.

The Healthy class (Table 7) is well-identified with 100% precision, 99.59% sensitivity, and 100% specificity, resulting in 99.89% accuracy. For the Common Rust class, the model exhibits excellent precision, sensitivity, and specificity of 96.10%, 99.19%, and 98.66%, respectively, achieving 98.79% accuracy. The Blight class shows reasonable precision of 93.78%, sensitivity of 95.58%, and specificity of 93.18%, resulting in 93.78% accuracy. The Gray Leaf Spot class demonstrates relatively good detection compared to previous algorithms, with sensitivity of 77.6%, specificity of 98.79%, precision of 95.57%, and 93.48% accuracy.

Table 7: evaluation metrics table for ResNet50

Classe	TP	TN	FN	FP	ACC (%)	SEN (%)	SPE (%)	PRE (%)	F1 Score (%)
Blight	238	697	11	51	93.78	95.58	93.18	82.35	90.17
Common Rust	247	738	2	10	98.79	99.19	98.66	96.10	97.62
Gray Leaf Spot	194	738	56	9	93.48	77.6	98.79	95.57	85.65
Healthy	248	747	1	0	99.89	99.59	100	100	99.80

• DenseNet201

In Figure 15, accuracy increases until epoch 6 (93.76% for training, 82.35% for testing) and stabilizes at epoch 6 (98.97% for training, 88.30% for testing). The training loss decreases rapidly until epoch 6 (3.24) and converges to 0.88.

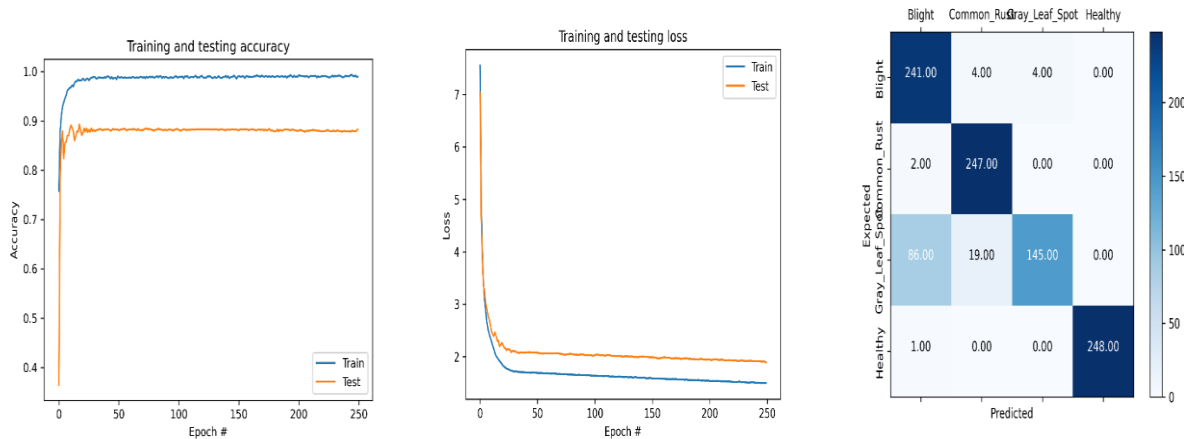


Figure 15 : Accuracy, loss, and confusion matrix of DenseNet.

The DenseNet model achieves good accuracy, with the Healthy class (Table 8) having excellent precision, sensitivity, and specificity (100%, 99.59%, and 100% respectively) with an accuracy of 99.89%. The Common Rust class demonstrates high precision, sensitivity, and specificity (92.16%, 99.19%, and 97.19% respectively) resulting in an accuracy of 97.69%. The Blight class exhibits reasonable precision (73.03%), sensitivity (96.78%), specificity (88.10%), and accuracy (90.27%).

Table 8: evaluation metrics table for ResNet50

Classe	TP	TN	FN	FP	ACC (%)	SEN (%)	SPE (%)	PRE (%)	F1 Score (%)
Blight	241	659	8	89	90.27	96.78	88.10	73.03	83.24
Common Rust	247	727	2	21	97.69	99.19	97.19	92.16	95.54
Gray Leaf Spot	145	743	105	4	89.06	58	99.46	97.31	72.68
Healthy	248	748	1	0	99.89	99.59	100	100	98.79

• MobileNet_V2

In Figure 16, accuracy increases until epoch 12 (97.26% for training, 81.45% for testing) and stabilizes at epoch 12 (99.64% for training, 91.43% for testing). The training loss decreases rapidly until epoch 12 (0.88) and then converges to 0.91.

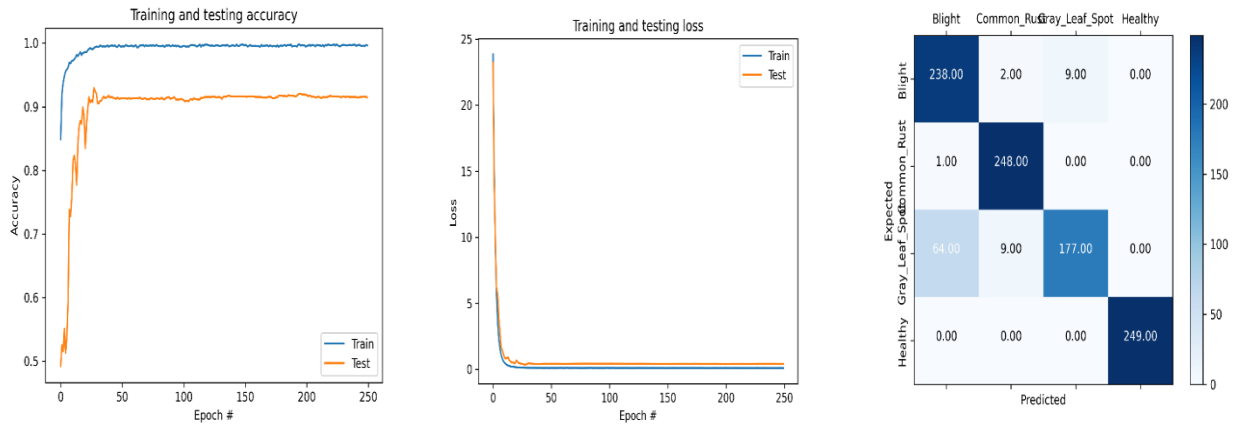


Figure 16 : Accuracy, loss, and confusion matrix of MobileNet.

The MobileNet_V2 model performs well, with the Healthy class (Table 9) having excellent precision, sensitivity, and specificity (100%, 100%, and 100% respectively) with an accuracy of 100%. The Common Rust class demonstrates high precision, sensitivity, and specificity (95.75%, 99.59%, and 97.79% respectively) resulting in an accuracy of 98.39%.

The Blight class exhibits reasonable precision (76.03%), sensitivity (95.58%), specificity (90.10%), and accuracy (91.45%). The Gray Leaf Spot class shows relatively good detection compared to other algorithms, with sensitivity at 70.8% and specificity and precision at 98.79% and 95.16% respectively. The accuracy for this class is 91.77%.

Table 9 : evaluation metrics table for MobileNet

Classe	TP	TN	FN	FP	ACC (%)	SEN (%)	SPE (%)	PRE (%)	F1 Score (%)
Blight	238	683	11	75	91.45	95.58	90.10	76.03	84.70
Common Rust	248	488	1	11	98.39	99.59	97.79	95.75	97.63
Gray Leaf Spot	177	738	73	9	91.77	70.8	98.79	95.16	81.20
Healthy	249	748	0	0	100	100	100	100	100

• Xception

Figure 17 shows that accuracy increases until epoch 15 (97.07% for training, 86.39% for testing) and stabilizes at epoch 15 (98.83% for training, 89.51% for testing). The training loss decreases rapidly until epoch 15 (0.59 for testing) and then converges to 0.44 for testing.

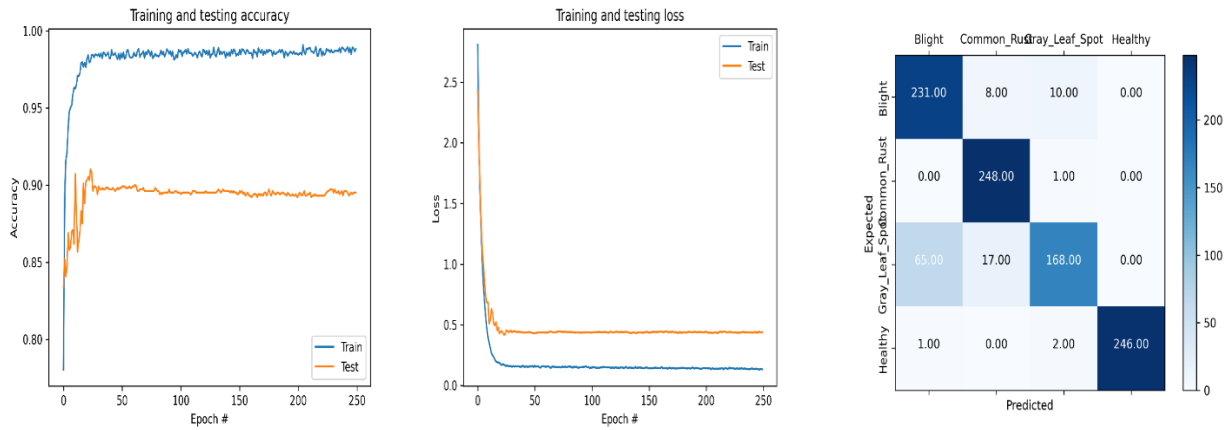


Figure 17 : Accuracy, loss, and confusion matrix of MobileNet.

The Xception model performs well, with the Healthy class (Table 10) exhibiting reasonable precision, sensitivity, and specificity (100%, 98%, and 100% respectively) with an accuracy of 99.89%. The Common Rust class demonstrates excellent precision, sensitivity, and specificity (90.84%, 99.59%, and 96.65% respectively) resulting in an accuracy of 97.39%.

The Blight class shows good detection with sensitivity and specificity at 92.77% and 91.17% respectively, along with a good precision of 77.78%. The accuracy for this class is 91.45%. The Gray Leaf Spot class is relatively well-detected compared to some other algorithms, with sensitivity at 67.2% and specificity and precision at 98.28% and 94.98% respectively. The accuracy for this class is 90.89%.

Table 10 : evaluation metrics table for MobileNet

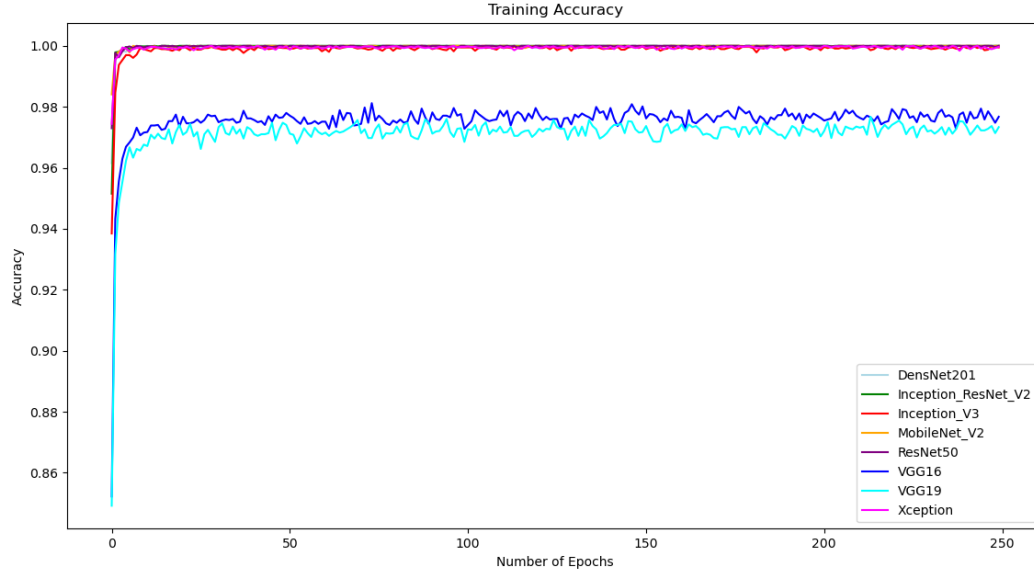
Classe	TP	TN	FN	FP	ACC (%)	SEN (%)	SPE (%)	PRE (%)	F1 Score (%)
Blight	231	682	18	66	91.57	92.77	91.17	77.78	84.61
Common Rust	248	723	1	25	97.39	99.59	96.65	90.84	85.01
Gray Leaf Spot	168	743	82	13	90.55	67.2	98.28	94.98	78.82
Healthy	246	748	3	0	99.89	98	100	100	98.99

4.8. Summary

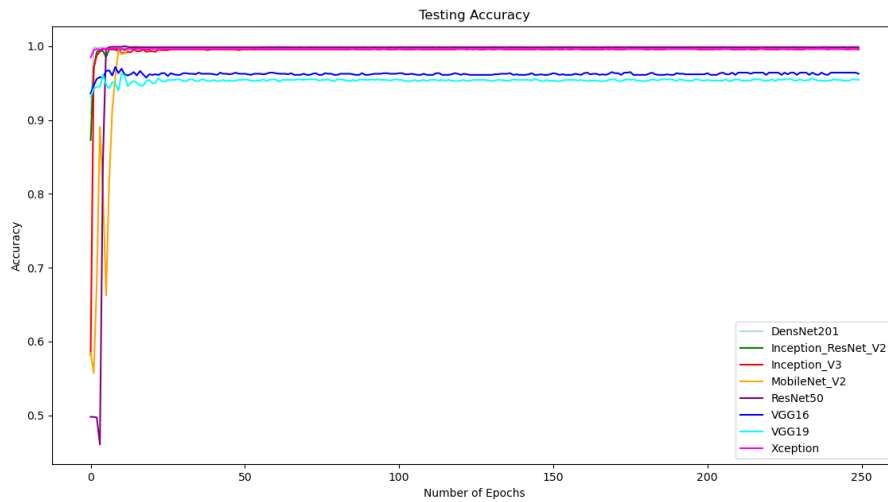
In this subsection, we give a summary of the performances of all models in binary and multiclass classification tasks.

4.8.1. Binary classification

The below figure 18, represents 2 plots of the training and testing accuracy curves of all trained models on the binary classification task.



(a)



(b)

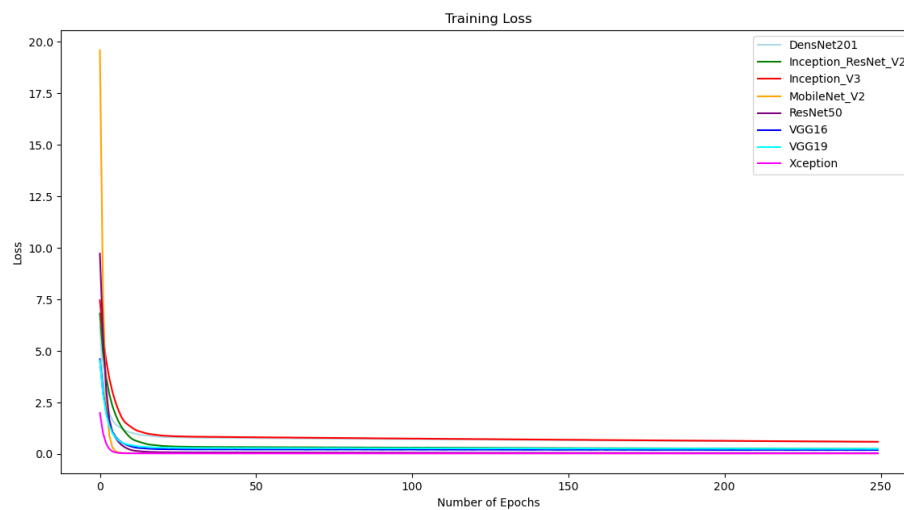
Figure 18: Summarization of the accuracy curves for different architectures.

For each model in Figure 18(a) and (b), which summarize the previous training and test accuracy figures, we observe that the training and test accuracy graphs increase until the point of stability. It is noted that the fine-tuned versions of Inception_Resnet_V2, Inception_V3, Resnet50, Densnet201, Xception, and Mobilenet_V2 demonstrate extremely satisfactory performances. Their accuracy

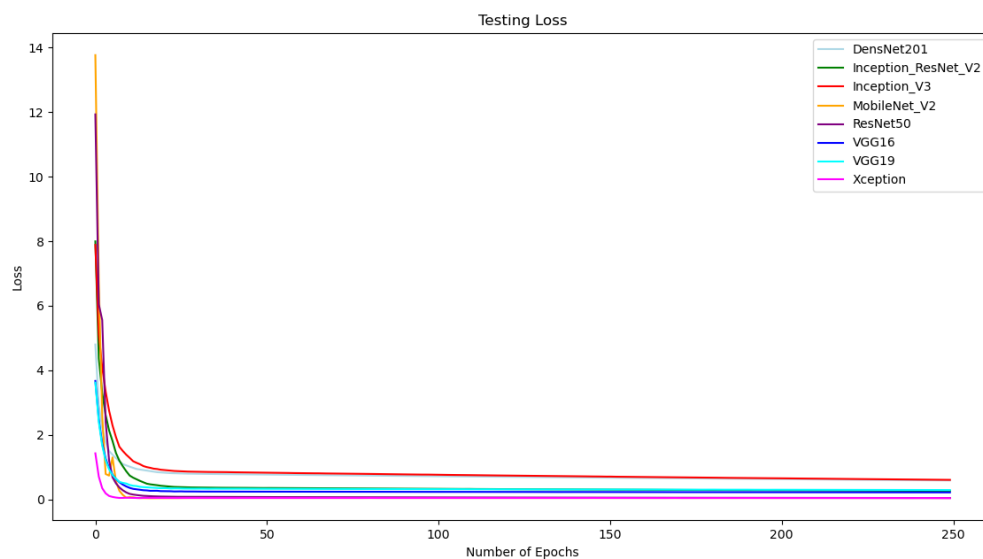
exceeds 99% for the training data, and their accuracy on the test data is respectively 99.70%, 99.54%, 99.85%, 100%, 99.70%, 99.70%. In particular, the Densnet201 algorithm achieves the maximum accuracy on the test data.

We observe also that the VGG architecture with its two versions, VGG16 and VGG19, achieved accuracy values of 96.28% and 95.44% respectively on the test data. These values are the lowest among the eight algorithms used in binary classification.

We can conclude that all the architectures trained on the binary classification problem have demonstrated very satisfactory performances. In particular, the DenseNet201 algorithm is the most performant model and has been chosen for production.



(a)



(b)

Figure19: Summarization of the loss curves for different architectures

For each model in Figure 19(a) and (b), which summarize the previous training and testing loss figures, we can notice that the training and testing loss graphs decrease exponentially until the point of stability. It is noteworthy that all the fine-tuned architectures demonstrate minimal and satisfactory loss values for both the training and test data. In particular the **MobileNet_V2** and **ResNet50** algorithms achieve minimal loss values on the test data close to 0.044.

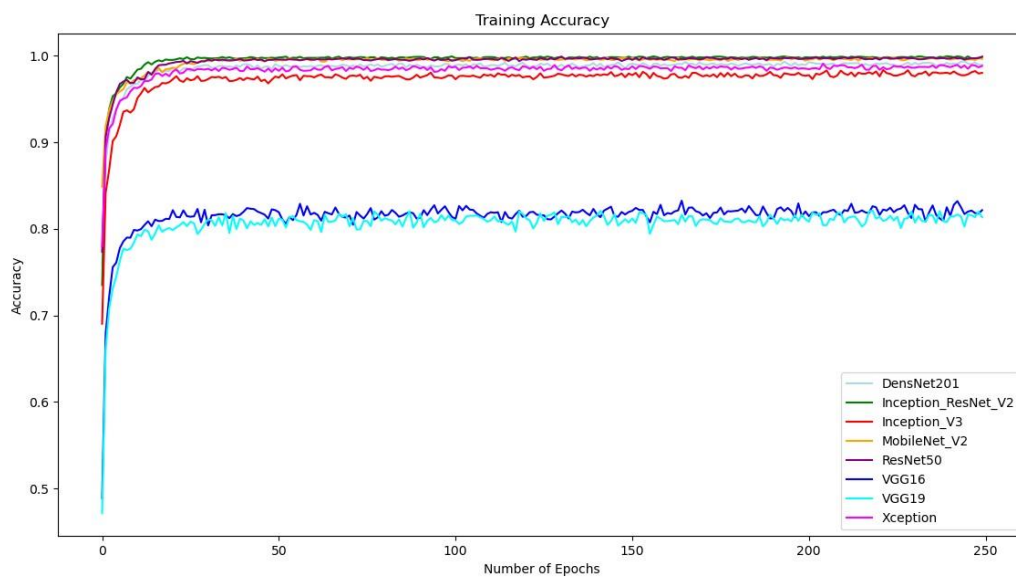
Tableau 3: Evaluation metrics in (%)

	TP	TN	FN	FP	ACC	SEN	SPE	PRE	F1
VGG16	653	614	6	43	96.28	99.03	93.45	93.82	96.35
VGG19	655	601	4	56	95.44	99.3	91.47	92.12	95.57
Xception	655	657	4	0	99.70	99.39	100	100	99.59
DenseNet201	659	657	0	0	100	100	100	100	100
Inception_V3	653	657	6	0	99.54	99.10	100	100	99.54
Inception_ResNet_V2	655	657	4	0	99.70	99.39	100	100	99.69
MobileNet_V2	656	657	3	0	99.70	99.55	100	100	99.77
ResNet50	647	657	2	0	99.85	99.70	100	100	99.84

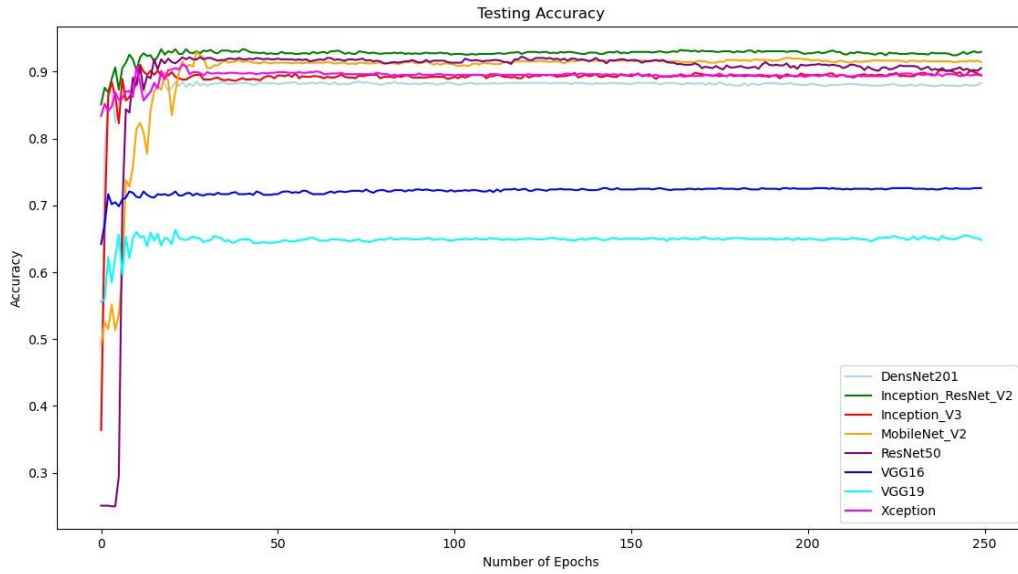
We can conclude that all the architectures trained on the binary classification problem have demonstrated very satisfactory performances (Table 11). In particular, the DenseNet201 algorithm is the most performing model, and it is indeed the algorithm chosen for production in the binary classification of corn leaf diseases.

4.8.2. Multiclass classification

The below figure 19, represents 2 plots of the training and testing accuracy curves of all trained models on the binary classification task.



(a)



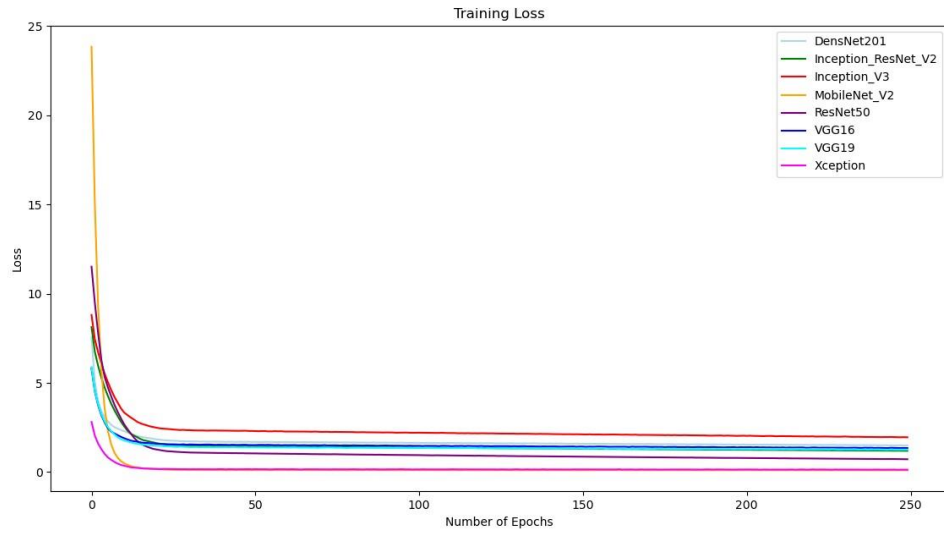
(b)

Figure 19 : Summarization of the accuracy curves for different architectures.

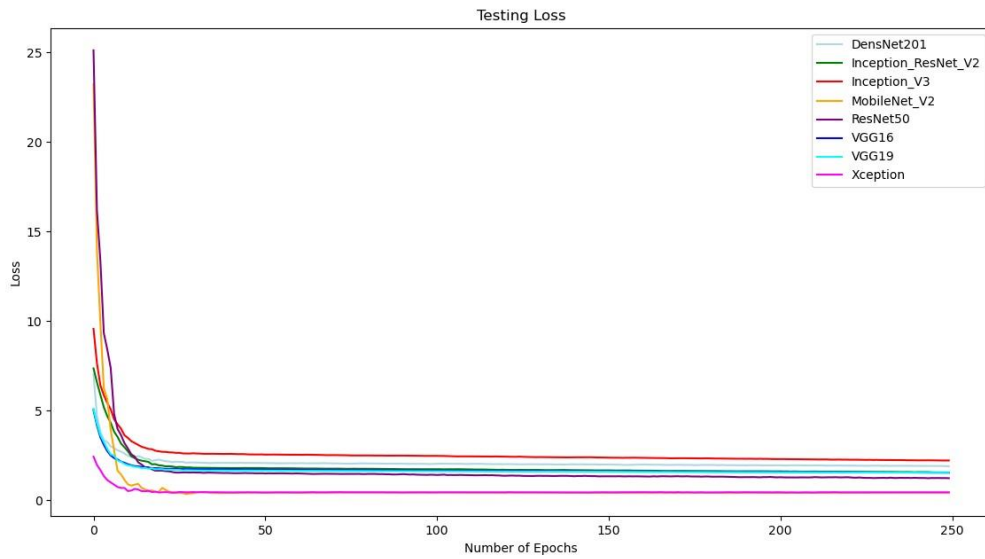
For each model in Figure 19(a) and (b), which summarize the previous training and testing accuracy figures, we see that the training and testing accuracy graphs increase until the point of stability. It is notable that the fine tuned versions of **Inception_Resnet_V2**, **Inception_V3**, **Resnet50**, **Densnet201**, **Xception**, and **Mobilenet_V2** demonstrate satisfactory performance, their accuracies exceed 98% for training data, while the accuracy for testing data hovers around 90% with accuracies of 92.98%, 89.46%, 90.57%, 88.36%, 89.56%, and 91.47%, respectively.

In particular, the **DensNet201** algorithm achieves maximum accuracy on the training data, recording an outstanding 99.9%, while **Inception_Resnet_V2** reaches a peak accuracy of 92.98% for the testing data.

However, the performance of the **VGG16** and **VGG19** algorithms proves relatively low, with accuracies around 80% for training data and not exceeding 70% for testing data. It is important to note that the **VGG19** algorithm shows the lowest accuracy among all the models studied, with a result of 64.99% for the testing data



(a)



(b)

Figure 20: Summarization of the loss curves for different architectures.

The training and testing loss graphs (Figure 20(a) and (b)) demonstrate a decrease in loss until reaching a point of stability for all models. The fine-tuned versions of the models show extremely satisfactory performance with low loss rates. It is interesting to note that the **MobileNet_V2** and **Xception** models stand out with the lowest loss values, not exceeding 0.13 and 0.45, respectively, for the training and testing data.

We can conclude that all the architectures trained for multi-class classification of maize leaf diseases have demonstrated very satisfactory performance, except for the **VGG16** and **VGG19** algorithms (Table 12)

Tableau 12 : Evaluation metrics in (%)

	ACC	SEN	SPE	PRE	F1
VGG16	72.71	75.26	90.92	75.89	69.12
VGG19	64.99	65.14	88.24	69.80	59.71
Xception	89.56	89.4	96.52	90.90	86.85
DenseNet201	88.36	88.40	96.19	90.62	87.56
Inception_V3	89.46	89.49	96.48	88.51	89.21
Inception_ResNet_V2	92.98	93.00	97.66	93.50	93.31
MobileNet_V2	91.47	91.50	96.68	91.74	90.88
ResNet50	90.57	90.60	96.86	91.27	90.48

The **Inception_Resnet_V2** algorithm has demonstrated to be the most performant model, making it the obvious choice for use in production. These observations clearly highlight the effectiveness of the models in the context of multi-class classification of corn leaf diseases.

5. Discussion

The results found in this study in which we scrutinize the performances of eight recent deep learning algorithms (**VGG16**, **VGG19**, **Inception_Resnet_V2**, **Inception_V3**, **Resnet50**, **Densnet201**, **Xception** et **MobileNet_V2**) form a solid basis to answer the research questions stated in the beginning.

5.1. (RQ1) What is (are) DL algorithm(s) that performs well on both binary and multiclass problems ?

The first research question focuses on the particular deep learning models that showed excellent performances in both binary and multiclass problems. The results obtained in our work revealed interesting insights concerning the eight algorithms implemented.

Based on the findings of this study about the classification of corn leaf disease images, we can confirm that the following algorithm: **Inception_Resnet_V2**, **Inception_V3**, **Resnet50**, **Densnet201**, **Xception**, and **MobileNet_V2** showed outstanding performances in both binary and multiclass classification tasks.

these conclusions can be explained by :

- **The complexity of the models:** The **Inception_ResNet_V2**, **DenseNet201**, **Inception_V3**, **ResNet50**, **Xception**, and **MobileNet_V2** architectures are deep and complex. These models employ advanced architectural designs that integrate various techniques to enhance performance.

- **InceptionResNet_V2** combines the advantages of the Inception and ResNet architectures by leveraging the power of deep residual learning and efficient feature extraction.
- **DenseNet201** make use of dense connections between layers, enabling a direct flow of information and improved gradient propagation.
- **MobileNet_V2** uses depthwise separable convolutions to achieve good balance between accuracy and computational efficiency.
- **Xception** also uses depthwise separable convolutions, placing more emphasis on capturing spatial relationships in the data.
- **Inception_V3**, an earlier version of the Inception models, introduced the concept of factorized convolutions to reduce computational complexity. This increase in depth enables a greater number of hierarchical and abstract feature representations

Furthermore, the use of these algorithms enables better capture and discrimination of the finest details in images. This is particularly important in the case of classifying corn leaf diseases, where different types of diseases may exhibit subtle variations. This is precisely the case with the two diseases, Blight and Gray Leaf Spot. The increased complexity of the model can be advantageous in distinguishing between them.

5.2.(RQ2) Does a high performing DL algorithm trained on a binary classification problem means that it will automatically achieve high performances on the multiclass problem as well ?

The second research question explores whether a high-performing DL algorithm trained on the binary classification problem will also achieve good performance as well in multi-class classification problem.

Our results indicate that a high performing DL algorithm trained on the binary classification does not mean by any way that this particular algorithm will show good performances on the multiclass task.

Based on our findings, the **VGG-net** architectures consisting of 2 versions; **VGG16** and **VGG19** have demonstrated excellent performances on the binary classification task, despite that they showed poor results in the multiclass task.

the previous conclusions can be justified by :

- **The influence of the algorithm's architecture:** Research conducted by Du et al [16], who carried out a comparative study of the performance of pre-trained models (AlexNet, VGG16, Inception, and ResNet), asserts that the topology of the classic single-channel CNN with coupled layers in the VGG architecture is less effective compared to the architectures used by Inception and ResNet models, which are respectively based on the Inception and Residual architectures.

these researches agree that non-linear architectures are much more suitable for classifying multiple classes of input images. In other words, the network's accuracy is improved when the categories of images vary, which is not the case for VGGNets.

- **The impact of transfer learning on the improvement of CNN architectures' performance:** Research has shown that new neural network architectures are better suited for transfer learning application, these models have achieved significant improvements in accuracy on the test set. This suggests that the new architectures are more capable of benefiting from transfer learning. In comparison, older architectures such as AlexNet and VGG are less influenced by transfer learning and generally show little improvement in accuracy [].

This observation confirms that the most complex Deep Learning architectures are capable of learning discriminative features from data, enabling them to effectively generalize to more complex classification scenarios.

Overall, our results highlight the relevance of the studied DL algorithms, such as **Inception_Resnet_V2**, **Inception_V3**, **Resnet50**, **Densenet201**, **Xception**, and **Mobilenet_V2**, for solving both binary and multi-class classification problems

Moreover, our findings support the idea that a high-performing Deep Learning algorithm in binary classification is generally capable of achieving good performance in multi-class classification task. However, it should be noted that this observation does not apply to older architectures with simple topology.

These findings support the application of these algorithms in real-world contexts and open up prospects for their use in a variety of classification applications in the field of precision agriculture.

6. Conclusions and future works

In this work, we presented automated methods for binary and multiclass classification using maize leaf disease images, by fine tuning eight deep learning architectures (**VGG16**, **VGG19**, **DenseNet201**, **Inception_ResNet_V2**, **Inception_V3**, **Xception**, **Resnet50**, and **MobileNet_V2**). The main goal was to answer the following research questions :

- (1) **What is (are) DL algorithm(s) that performs well on both binary and multiclass problems ?**
- (2) **Does a high performing DL algorithm trained on a binary classification problem means that it will automatically achieve high performances on the multiclass problem as well ?**

The experiments were conducted using corn leaf disease open dataset, which contains 4188 images. Additionally, the performances of these experiments were evaluated using various performance metrics. Moreover, the results indicate that DenseNet201 showed the highest performance in the binary classification task, on the hand Inception_Ren_Net_V2 achieved the best performance on the multiclass classification task.

References

- [1] El Asnaoui, K., Chawki, Y., & Idri, A. (2020). Automated methods for detection and classification. *Artificial Intelligence and Blockchain for Future Cybersecurity Applications*. https://doi.org/10.1007/978-3-030-74575-2_14.
- [2] El Asnaoui, K., & Chawki, Y., (2020). Using X-ray images and deep learning for automated detection of coronavirus disease. *Journal of Biomolecular Structure and Dynamics*. <https://doi.org/10.1080/07391102.2020.1767212>.
- [3] Singh, D., Jain, N., Jain, P., Kayal, P., Kumawat, S., & Batra, N. (2020). PlantDoc: a dataset for visual plant disease detection. In *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD 2020*.
- [4] Arun Pandian, J., & Gopal, Geetharamani. (2019). Data for: Identification of Plant Leaf Diseases Using a 9-layer Deep Convolutional Neural Network.
- [5] Padilla, D. A., Pajes, R. A. I., & Guzman, J. T. D. (2020). Detection of Corn Leaf Diseases Using Convolutional Neural Network with OpenMP Implementation.
- [6] Xu, Y., Zhao, B., Zhai, Y., Chen, Q., & Zhou, Y. (2022). Maize Diseases Identification Method Based on Multi-Scale Convolutional Global Pooling Neural Network. *IEEE Access*, 9, 27959-27970. <https://doi.org/10.1109/ACCESS.2021.3058267>.
- [7] Alom, M. Z., Taha, T. M., Yakopcic, C., Westberg, S., Sidike, P., Nasrin, M. S., Van Essen, B. C., Awwal, A. A. S., & Asari, V. K. (2018). The history began from alexnet: a comprehensive survey on deep learning approaches.
- [8] Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition.
- [9] Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016). Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 2818-2826).
- [10] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 770-778).
- [11] Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 4700-4708).
- [12] Huang, G., Liu, Z., Van Der Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4700–4708 (2017).
- [13] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). MobileNetV2: inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 4510-4520).
- [14] Chollet, F. (2017). Xception: deep learning with depthwise separable convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 1251-1258).
- [15] Smaranjit, G. (2020). Corn or Maize Leaf Disease Dataset. Retrieved March 1, 2023, from <https://www.kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf-disease-dataset>.

- [16] Du, X., Sun, Y., Song, Y., Sun, H., & Yang, L. (2023). A Comparative Study of Different CNN Models and Transfer Learning Effect for Underwater Object Classification in Side-Scan Sonar Images. *Remote Sensing*, 15, 593. <https://doi.org/10.3390/rs15030593>.